

Additive oxygen effects in Cl₂ plasma etching of chrome films

KWANG-HO KWON

Department of Electronic Engineering, Hanseo University, 360, Daegok-Ri, Haemi-Myun, Seosan-Si, Chung-Nam, 356-820, Korea
E-mail: khkwon@gaya.hanseo.ac.kr

SEUNG-YOUL KANG, SANG-HO PARK, HEE-KYUNG SUNG

Electronics and Telecommunication Research Institute, Yusong P.O. Box 106, Taejon 305-600, Korea

DONG-KEUN KIM*, JONG-HA MOON†

*Department of *Metallurgical Engineering and †Ceramic Engineering, Chonnam National University, Kwangju, 500-757, Korea*

Chrome films have been used in many applications such as field emission display (FED) gate electrodes [1], mask patterns of photolithography, and thin film transistor (TFT) liquid crystal display (LCD) electrodes [2, 3]. Since FED and TFT LCD devices are especially becoming more widely used, we will continue to see Cr films used in more device fabrications. In addition, Cr films are normally used for mask patterns due to their good reflectivity of light and low coefficient of thermal-linear expansion ($\alpha = 4.9 \times 10^{-6} [\text{K}^{-1}]$) [4]. In the past, wet Cr etching for the mask patterns with solutions which consist of ceric ammonium nitrate ($\text{Ce}(\text{NH}_4)_2\text{NO}_3)_6$), perchloric acid (NClO_4 (70%)), and deionized (DI) water was mainly carried out. However, the decrease of integrated circuit (IC) design rule requires the finer pattern formation on a quartz plate. This limits the usage of wet Cr etching. At the same time, the pattern size of Cr gate in the FED devices has to be defined under the submicron. On the other hand, the wet etching has a reduction of pattern size. In order to overcome this drawback of the wet Cr etching, reactive ion etching (RIE) of Cr films is a promising candidate for the formation technique of submicron pattern due to the property of precise pattern transfer. Therefore, dry etching is necessary for the etching of Cr films. However, dry etching of Cr films has been scarcely examined until now. At the same time, to improve the etching characteristics, the surface reaction between the Cr surface and plasma species has to be closely comprehended. In this study, we etched Cr films by Cl_2/O_2 plasmas, and the surface reaction of Cr etching was investigated by using X-ray photoelectron spectroscopy (XPS). The role of the additive O_2 gas and the relationship between the etch rate and the gas-mixing ratio will be further discussed in this article.

The 5-in. Si substrate used for this study was doped with B ($0.85\text{--}1.15 \, \Omega\text{-cm}$), oriented (100) and chemically etched for 60 s using 1% HF : H_2O prior to chemical vapor deposition (CVD) growth. The substrate was deposited with a 600-nm-thick layer of SiO_2 grown by low-pressure-CVD ($\text{SiH}_4 + \text{O}_2$, 420°C , 240 mTorr). The deposition of the Cr films was performed using a Varian 3180 dc sputtering system and was described in the previous work in detail [5]. The Cr sputtering target

was specified at 99.99% purity. The final thickness of the sputtered Cr film was ~ 400 nm. The mask for the Cr etch was a silicon oxide layer which was deposited by electron cyclotron resonance (ECR) CVD method. The source gases were SiF_4 (40 sccm) and N_2O (80 sccm) and the microwave power and the process pressure were 700 W and 20 mTorr, respectively. The process was carried out at room temperature. The oxide deposited by this method involved a few atomic percentages of fluorine.

Reactive ion etching of the Cr films was performed using a Plasma Therm 790 RIE etching system. Wafers were placed on a bottom electrode. The temperature of substrate holder during the Cr etch was held at $\sim 20^\circ\text{C}$ using the circulation of cooling water. The Cl_2/O_2 gas mixture was used for the etch, and the total gas flow was constant and 40 sccm. The $\text{O}_2/(\text{O}_2 + \text{Cl}_2)$ mixing ratio was changed from 0 to 0.5. RF power and chamber pressure were constant at 350 W and 150 mTorr, respectively. The vertical etch rate was measured through sheet resistance using a four-point probe and Hitachi SEM (scanning electron microscope). And the lateral etch rate was measured using SEM.

Compositional analysis of the Cr surface was performed using a VG Scientific ESCALAB 200R XPS (X-ray photoelectron spectroscopy) with AlK_{α} (1486.6 eV) radiation operating at 260 W. Narrow scan spectra of all regions of interest were recorded with 20 eV pass energy to quantify the surface composition and identify the chemical binding states.

The surface reaction of Cr etched with Cl_2/O_2 -gas plasma was investigated by using XPS analysis. Fig. 1 represents the atomic percentages of Cr, Cl, O and C with various Cl_2/O_2 gas-mixing ratios. This figure shows that the atomic percentages of O increase but those of Cl decrease with increasing the ratio. This seems to result from the increased partial pressure of O_2 in the reaction chamber. That is, the increased partial pressure of O_2 induces the increase of O species and the decrease of Cl species in the reaction chamber. At the same time, the atomic percentages of Cr also increase with increasing the ratio. Meanwhile, carbon atoms detected on the sample seem to result from an air exposure of the etched surface.

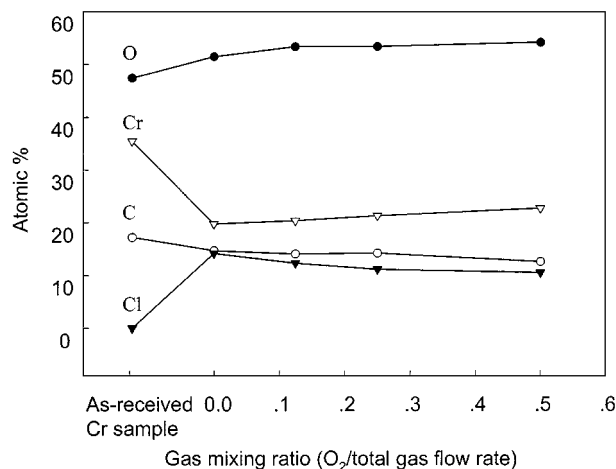


Figure 1 The changes of at % with various gas-mixing ratios.

The XPS narrow scan analysis was carried out to examine the cause of atomic percentage change in Fig. 1. Fig. 2 represents that the narrow scan spectra of Cr 2p with various Cl_2/O_2 gas-mixing ratios. In this figure, the four peaks in the as-received Cr sample were observed. Cr 2p_{3/2} and Cr 2p_{1/2} peaks from elementary Cr occurred at the binding energies of 574.4 and 583.8 eV, respectively. These elementary Cr peaks were confirmed by using the Cr 2p spectra of Ar-sputtered Cr sample. In the Cl_2 etched sample, other peaks were also observed at the binding energy of 578 and 587.6 eV. It is generally known that the Cr 2p_{3/2} peak at the binding energy is formed from Cr–Cl bond. The peaks at the binding energies of 577 and 586.6 eV are also observed in the as-received sample. These peaks come from typical Cr–O bonds of Cr_2O_3 compounds [6]. This figure indicates that the additive O_2 gas makes peaks move to lower binding energy. This phenomenon implies that a chromium atom combines with chlorine and oxygen atoms to form Cr–Cl and Cr–O bonds on the Cr surface.

Fig. 3 shows the Cl 2p narrow scan spectra with the ratio. From the figure, a peak at the binding energy of 197.9 eV was observed and came from typical metal–Cl bonds. Generally, the Cl 2p peak consists of Cl 2p_{3/2} and Cl 2p_{1/2} peaks, of which the difference of binding energies is 1.6 eV. This figure implies that the Cl species on the Cr surface mainly bind to Cr. The shape and the binding energy changes of Cl 2p spectra are not observed with increasing the ratio. This means that elements bonded to Cl atoms are not changed in spite of the O_2 addition. That is, the O_2 addition into Cl_2 plasma does not have an effect on Cl bond states.

Fig. 4 displays the O 1s narrow scan spectra with the ratio. In the Ar-sputtered sample, the maximal peak position was observed at the binding energy of 531.3 eV. This peak comes from typical metal–oxygen bonds, that is, Cr–O bonds from Cr_2O_3 . At the same time, it was found that the spectra of O 1s in the Cl_2/O_2 -etched sample and as-received Cr sample were nearly same. This indicates that oxygen combines with Cr in spite of the existence of Cl_2 gas in the plasma. This phenomenon agrees to the fact that oxygen atoms hardly combine with chlorine atoms due to high electronega-

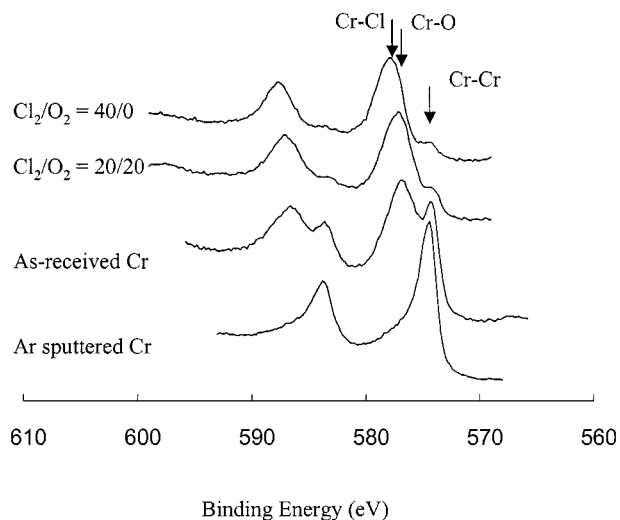


Figure 2 Cr 2p spectra with various Cl_2/O_2 gas-mixing ratios.

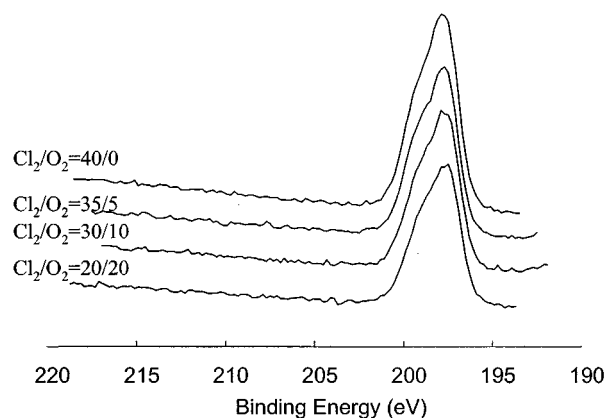


Figure 3 Cl 2p spectra with various gas-mixing ratios.

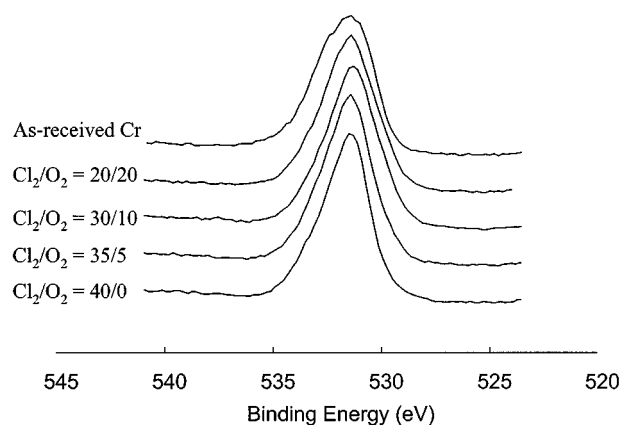


Figure 4 O 1s spectra with gas-mixing ratios.

tivities of chlorine and oxygen. From Fig. 3, this implies the existence of CrO_xCl_y compounds (O_x -Cr- Cl_y) on the etched Cr surface.

To investigate the cause of Cr etch rate with the ratio, plasma characteristics were also examined. Fig. 5 represents the V_{dc} (self-bias) with the ratio. This figure shows that the dc-self bias voltage increases with increasing the ratio above 0.13. However, at the ratios of 0.13 and below, the dc-self bias voltage is nearly

constant. This indicates that ion bombardment effects on the etched surface increase with increasing the ratio. So, the etch rate of Cr films seems to be largely affected by the ratio.

Fig. 6 plots the Cr etch rate with increasing the ratio. The lateral etch rate was calculated by using SEM images of etched Cr patterns. This figure shows the increase of Cr etch rate with increasing the ratio. It was found in this figure that at the ratios of 0.07 and over, the Cr etch rate linearly increased with increasing the ratio and abruptly increased at the ratio of 0.07. This trend does not agree to the changes of dc-self bias with increasing the ratio of 0.13 and below. That is, the Cr etch rate abruptly increased in spite of nearly the same dc-self bias voltage at the ratios of 0.13 and below. This can be explained as follows: When Cr is etched by using Cl-based gas plasma, CrCl_n ($n = 1-3$) compounds are likely to be formed and exist on the surface. CrCl_n compounds are not easily removed on the etched surface due to the low vapor pressure of Cr-related etch products such as CrCl_3 in Cl-based etching. This can be inferred from its high melting point of 1152°C [7]. Meanwhile, the addition of O_2 gas in the Cl_2 -gas plasma results in the formation of CrO_xCl_y ($x = 1-2$, $y = 1-2$) compounds on the etched Cr surface. The existence of CrO_xCl_y compounds is confirmed in Fig. 2. The CrO_xCl_y compounds are easily removed on the etched surface due to the high vapor pressure of Cr-related etch products such as CrO_2Cl_2 . In the case of CrO_2Cl_2 compound, its vapor pressure is very high and 1 mm Hg at the temperature of -18.4°C . So, Cr etch rate increases with increasing the ratio in spite of a little change of dc-self bias voltage at the low ratios. At the same time, the lateral etch rate linearly increased with increasing the ratio. It was generally known that the lateral etch was progressed by chemical reaction and the removal of by-products on the etched surface. Therefore, this change of the lateral etch rate also implies the formation of volatile compounds such as CrO_xCl_y on the etched surface. Meanwhile, the slope of the vertical etch rate in the Cr etching at the ratios of 0.07 and over is steeper than that of the lateral etch rate. This difference seems to result from the increase of dc-self bias voltage with the ratio. From this point of view, the increase of Cr etch rate with the ratio results from the increase of dc-self bias voltage and the formation of CrO_xCl_y compounds on the etched surface.

Fig. 7 represents the Cr profile etched by using an oxide mask pattern at the ratio of 38/2. This sample was overetched by the 30% of Cr film thickness. This figure shows non-lateral etch and vertical profile in spite of 30% overetching.

A Cr surface was characterized by XPS after Cl_2/O_2 -gas plasma etching. The relationship between the etch rate and the gas-mixing ratio was investigated. The relationship between the etched profile and the ratio was also investigated. In the Cl_2 etched sample, Cr 2p peaks from Cr-Cl bond were observed. At the same time, the additive O_2 gas made peaks move into lower binding energy. This phenomenon indicates that Cr combines chlorine and oxygen to form Cr-Cl and Cr-O bonds on the Cr surface. And, it was confirmed from the ex-

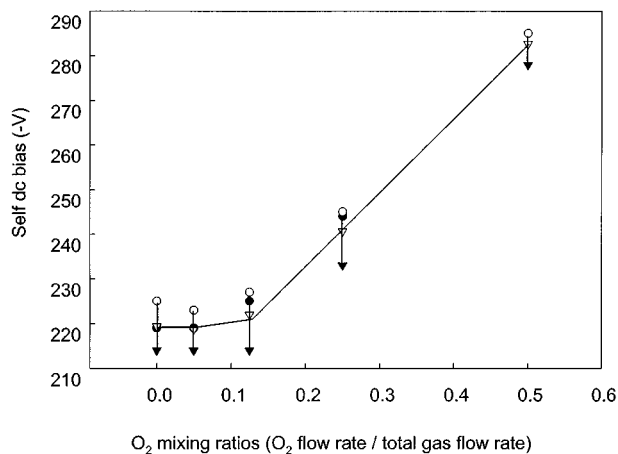


Figure 5 V_{dc} variation with various gas-mixing ratios.

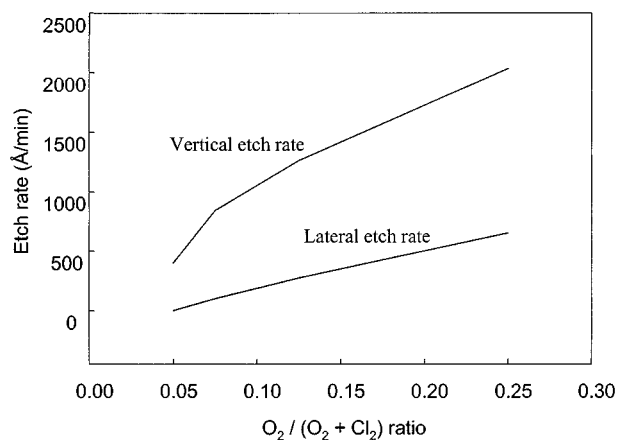


Figure 6 The Cr etch rate with increasing the gas-mixing ratios.

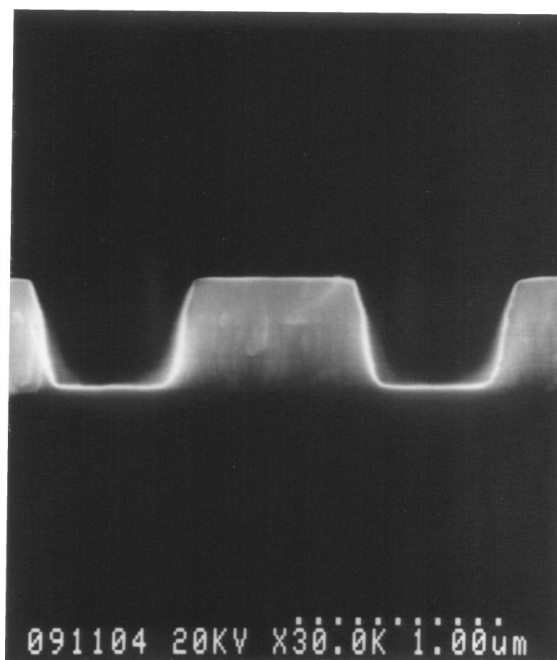


Figure 7 SEM image of etched Cr profile at the Cl_2/O_2 ratio of 38/2 sccm.

amination of O 1s and Cl 2p narrow scan spectra that Cl and O species on the Cr surface mainly bound to Cr. This implies the existence of CrO_xCl_y compounds on the etched Cr surface. On the other hand, it was

found that the Cr etch rate abruptly increased in spite of nearly the same dc-self bias at the low ratio. It can be inferred that the abrupt increase of Cr etch rate is related to the formation of the CrO_xCl_y compounds from additive O_2 and results from the high vapor pressure of the compounds. Meanwhile, the slope of the vertical etch rate in the Cr etching at the high ratios was higher than that of the lateral etch rate and this difference seems to result from the increase of dc-self bias voltage with the ratio. From this point of view, the increase of Cr etch rate with the ratio results from the increase of dc-self bias voltage and the formation of CrO_xCl_y compounds on the etched surface.

Acknowledgment

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References

1. S. E. HUQ, M. HUANG, P. R. WILSHAW and P. D. PREWETT, in The Tenth International Vacuum Microelectronics Conference, 1997, p. 412.
2. JUN-WEI TSAI, CHUN-YAO HUANG, YA-HSIANG TAI, HUANG-CHANG CHENG, FENG-CHENG SU, FANG-CHEN LUO and HSING-CHEN TUAN, *Appl. Phys. Lett.* **71** (1997) 1237.
3. CHUN-SUNG CHIANG, CHUN-YING CHEN, JERZY KANICKI and KAZUSHIGE TAKECHI, *ibid.* **72** (1998) 2874.
4. DAVID R. LIDE (ed.), "CRC Handbook of Chemistry and Physics," 74th ed., 1993, pp. 12-135.
5. K.-H. BAEK, S. J. YUN, J.-M. PARK, Y.-S. YOON, K.-S. NAM, K.-H. KWON and C.-I. KIM, *J. Mater. Sci. Lett.* **17** (1998) 1483.
6. J. F. MOULDER, W. F. STICKLE, P. E. SOBOL and K. D. BOMBEN (eds.), "Handbook of X-ray Photoelectron Spectroscopy" (Perkin-Elmer, 1992) p. 76.
7. M. WINDHOLZ (ed.), "The Merck Index," 9th ed., 1976, p. 287.

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